



Angeliq®

Film-Coated Tablet

1. NAME OF THE MEDICINAL PRODUCT

Angeliq 1 mg/2 mg film-coated tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Angeliq-1 mg E2 / 2 mg DRSP: Each medium red film-coated tablet contains: estradiol (E2) 1.0 mg (as estradiol hemihydrate), drospirenone (DRSP) 2.0 mg

3. PHARMACEUTICAL FORM

Film-coated tablet

4. CLINICAL PARTICULARS

4.1 Indication(s)

Hormone replacement therapy (HRT) for the treatment of:

- Climacteric syndrome in postmenopausal women including vasomotor symptoms (such as hot flushes and sweating attacks), sleep disorders, depressed moods, nervousness and atrophic urogenital conditions caused by deficient endogenous estrogen production due to natural menopause, hypogonadism, castration or primary ovarian failure in women with an intact uterus.
- Prevention of postmenopausal osteoporosis.

4.2 Dosage and method of administration

4.2.1 Method of administration

Oral use

4.2.2 Dosage regimen

► How to start Angeliq

Women who do not take estrogens or women who change from a continuous combination product may start treatment at any time.

Women changing from a continuous sequential or cyclic HRT should complete the current cycle of therapy before initiating Angeliq therapy.

► Dosage

One tablet is taken daily.

► Administration

Each pack covers 28 days of treatment. Treatment is continuous, which means that the next pack follows immediately without a break.

The tablets are to be swallowed whole with some liquid irrespective of food intake.

The tablets should preferably be taken at the same time every day.

For treatment of postmenopausal symptoms, the lowest effective dose should be used.

For initiation and continuation of treatment of postmenopausal symptoms, the lowest effective dose for the shortest duration should be used.

► Missed tablets

In case a tablet is forgotten, it should be taken as soon as possible. If more than 24 hours have elapsed, no extra tablet needs to be taken. If several tablets are forgotten, bleeding may occur.

4.2.3 Additional information on special populations

4.2.3.1 Pediatric patients

Angeliq is not indicated for use in children and adolescents.

4.2.3.2 Geriatric patients

There are no data suggesting a need for dosage adjustment in elderly patients.

4.2.3.3 Patients with hepatic impairment

In women with mild or moderate hepatic impairment, DRSP is well tolerated (see section 'Pharmacokinetic properties'). Angeliq is contraindicated in women with presence or history of liver tumors and with severe hepatic disease (see section 'Contraindications'). For women with impaired liver function, close supervision is needed and in case of deterioration of markers of liver function, use of HRT should be stopped (see section 'Special warnings and precautions for use').

4.2.3.4 Patients with renal impairment

In women with mild or moderate renal impairment, a slight increase of DRSP exposure was observed but is not expected to be of clinical relevance (see section 'Pharmacokinetic properties'). Angeliq is contraindicated in women with severe renal disease (see section 'Contraindications').

4.3 Contraindications

Hormone replacement therapy (HRT) should not be started in the presence of any of the conditions listed below. Should any of the conditions appear during HRT use, the product should be stopped immediately.

- ▶ Undiagnosed abnormal genital bleeding
- ▶ Known or suspected cancer of the breast
- ▶ Known or suspected premalignant conditions or malignancies, if sex steroid-influenced
- ▶ Presence or history of liver tumors (benign or malignant)
- ▶ Severe hepatic disease
- ▶ Presence or history of severe renal disease as long as renal function values have not returned to normal
- ▶ Acute arterial thromboembolism (e.g. myocardial infarction, stroke)
- ▶ Active deep venous thrombosis, thromboembolic disorders, or a documented history of these conditions
- ▶ A high risk of venous or arterial thrombosis
- ▶ Severe hypertriglyceridemia
- ▶ Pregnancy or lactation (see section 'Pregnancy and lactation')
- ▶ Hypersensitivity to the active substances or to any of the excipients

4.4 Special warnings and precautions for use

Before initiating therapy, all conditions/risk factors mentioned below should be considered when determining the individual benefit/risk of treatment for the patient.

During HRT use, **therapy should be discontinued immediately** in case a contraindication is discovered, as well as in the following situations:

- ▷ Migrainous or frequent and unusually severe headaches that occur for the first time or other symptoms that are possible prodroma of cerebrovascular occlusion.
- ▷ Recurrence of cholestatic jaundice or cholestatic pruritus which occurred first during pregnancy or previous use of sex steroids.
- ▷ Symptoms of a thrombotic event or suspicion thereof.

In the event of new onset or deterioration of the following conditions or risk factors, the individual benefit/risk analysis should be re-done, taking into consideration the possible necessity of discontinuing therapy.

The potential for an increased synergistic risk of thrombosis should be considered in women who possess a combination of risk factors or exhibit a greater severity of an individual risk factor. This increased risk may be greater than a simple cumulative risk of the factors. HRT should not be prescribed in case of a negative risk benefit assessment.

▶ Venous thromboembolism

Both randomized-controlled and epidemiological studies have suggested an increased relative risk (RR) of developing venous thromboembolism (VTE), i.e. deep venous thrombosis or pulmonary embolism. Benefit/Risk should therefore be carefully weighed in consultation with the patient when prescribing HRT to women with a risk factor for VTE.

Generally recognized risk factors for VTE include a personal history, a family history (the occurrence of VTE in a direct relative at a relatively early age may indicate genetic disposition) and severe obesity. The risk of VTE also increases with age. There is no consensus about the possible role of varicose veins in VTE.

The risk of VTE may be temporarily increased with prolonged immobilization, major elective or post-traumatic surgery, or major trauma. Depending on the nature of the event and the duration of the immobilization, consideration should be given to a temporary discontinuation of HRT.

► Arterial thromboembolism

Two large clinical trials with continuous combined conjugated equine estrogens (CEE) and medroxyprogesterone acetate (MPA) showed a possible increased risk of coronary heart disease (CHD) in the first year of use and no benefit thereafter. One large clinical trial with CEE alone showed a potential reduction of CHD rates in women aged 50-59 and no overall benefit in the total study population. As a secondary outcome, in two large clinical trials with CEE alone or combined with MPA a 30-40% increased risk of stroke was found. It is uncertain whether these findings also extend to other HRT products or non-oral routes of administration.

► Gallbladder disease

Estrogens are known to increase the lithogenicity of the bile. Some women are predisposed to gallbladder disease during estrogen therapy.

► Dementia

There is limited evidence from clinical studies with CEE-containing preparations that hormonal treatment may increase the risk of probable dementia if initiated in women aged 65 or older. The risk may be decreased if treatment is initiated in the early menopause, as observed in other studies. It is unknown whether these findings also extend to other HRT products.

Tumors

► Breast cancer

Clinical and observational studies have reported an increased risk of having breast cancer diagnosed in women taking HRT for several years.

Estimates for the overall relative risks of breast cancer diagnosis given in more than 50 epidemiological studies ranged in the majority of the studies between 1 and 2.

The relative risk increases with duration of treatment and may be lower or possibly neutral with estrogen-only products. Two large randomized trials with CEE alone or continuously combined with MPA showed risk estimates of 0.77 (95% confidence interval (CI): 0.59-1.01) or 1.24 (95%CI: 1.01-1.54) after approximately 6 years of HRT use. It is unknown whether the increased risk also extends to other HRT products.

The excess risk decreases within a few years after stopping HRT.

HRT increases the density of mammographic images which may adversely affect the radiological detection of breast cancer in some cases.

► Ovarian Cancer

A meta-analysis from 52 epidemiological studies reported that the overall risk of being diagnosed with ovarian cancer is slightly increased for users of HRT compared to women who have never used HRT (prospective studies: RR 1.20, 95% CI 1.15-1.26; all studies combined: RR 1.14, 95% CI 1.10-1.19). In women currently using HRT the risk of ovarian cancer was further increased (RR 1.43, 95% CI 1.31-1.56).

These associations have not been shown in all studies including randomised controlled trials, e.g. the Women's Health Initiative (WHI).

Furthermore, an effect of duration of exposure has not been consistently shown, but the risk may be more relevant with long-term use (several years).

► Endometrial cancer

Prolonged exposure to unopposed estrogens increases the risk of development of endometrial hyperplasia or carcinoma. The addition of DRSP opposes the development of endometrial hyperplasia caused by estrogens.

► Liver tumors

In rare cases benign, and even more rarely, malignant liver tumors have been observed after the use of hormonal substances such as those contained in HRT products. In isolated cases, these tumors led to life-threatening intra-abdominal hemorrhage.

Other conditions

A general association between HRT use and development of clinical hypertension has not been established. Small increases in blood pressure have been reported in women taking HRT, clinically relevant increases are rare. However, if in individual cases a sustained clinically significant hypertension develops during the use of HRT then withdrawing the HRT may be considered.

Angeliq has the potential to lower blood pressure in women with elevated blood pressure. In normotensive women, relevant changes in blood pressure are not expected.

Potassium excretion capacity may be limited in patients with renal insufficiency. In a clinical study, DRSP intake did not show an effect on the serum potassium concentration in patients with mild or moderate renal impairment. A theoretical risk for hyperkalemia can be assumed only for patients whose pretreatment serum potassium is in the upper reference range, and who are additionally using potassium sparing drugs.

Non-severe disturbances of liver function, including hyperbilirubinemias such as Dubin-Johnson syndrome or Rotor syndrome, need close supervision and liver function should be checked periodically. In case of deterioration of markers of liver function, use of HRT should be stopped.

Women with moderately elevated levels of triglycerides need special surveillance. HRT in these women may be associated with a further increase in triglyceride levels bearing the risk of acute pancreatitis.

Although HRT may have an effect on peripheral insulin resistance and glucose tolerance, there is generally no need to alter the therapeutic regimen in diabetics using HRT. However, diabetic women should be carefully monitored while taking HRT.

Certain patients may develop undesirable manifestations of estrogenic stimulation under HRT such as abnormal uterine bleeding. Frequent or persistent abnormal uterine bleeding during treatment is an indication for endometrial assessment.

Uterine fibroids (myomas) may increase in size under the influence of estrogens. If this is observed, treatment should be discontinued.

Should endometriosis be reactivated under treatment, discontinuation of therapy is recommended.

Close medical supervision (including periodic measurement of prolactin levels) is necessary if the patient suffers from prolactinoma.

Chloasma may occasionally occur, especially in women with a history of chloasma gravidarum. Women with a tendency to chloasma should avoid exposure to the sun or ultraviolet radiation whilst taking HRT.

The following conditions have been reported to occur or deteriorate with HRT use. Although the evidence of an association with HRT use is inconclusive, women with these conditions and treated with HRT should be carefully monitored.

Epilepsy
Benign breast disease
Asthma
Migraine
Porphyria
Otosclerosis
Systemic lupus erythematosus
Chorea minor

In women with hereditary angioedema exogenous estrogens may induce or exacerbate symptoms of angioedema.

4.5 Interaction with other medicinal products and other forms of interaction

► Effects of other medicinal products on Angeliq

Interactions can occur with drugs that induce microsomal enzymes which can result in increased clearance of sex hormones and which may lead to changes in the uterine bleeding profile and/or reduction of the therapeutic effect. Enzyme induction can already be observed after a few days of treatment. Maximal enzyme induction is generally seen within a few weeks. After the cessation of drug therapy enzyme induction may be sustained for about 4 weeks.

Substances increasing the clearance of sex hormones (diminished efficacy by enzyme-induction), e.g.:

Phenytoin, barbiturates, primidone, carbamazepine, rifampicin, and possibly also oxcarbazepine, topiramate, felbamate, griseofulvin and products containing St. John's wort.

Substances with variable effects on the clearance of sex hormones.:

When co-administered with sex hormones, many HIV/HCV protease inhibitors and non-nucleoside reverse transcriptase inhibitors can increase or decrease plasma concentrations of estrogen or progestin or both. These changes may be clinically relevant in some cases.

Substances decreasing the clearance of sex hormones (enzyme inhibitors)

Strong and moderate CYP3A4 inhibitors such as azole antifungals (e.g. fluconazole, itraconazole, ketoconazole, voriconazole), verapamil, macrolides (e.g. clarithromycin, erythromycin), diltiazem and grapefruit juice can increase plasma concentrations of the progestin or the estrogen or both.

Interaction with alcohol

Acute alcohol ingestion during use of HRT may lead to elevations of circulating E2 levels.

► Effect of Angeliq on other medicinal products

Based on in vivo interaction studies in female volunteers using omeprazole, simvastatin, or midazolam as marker substrates, a clinically relevant interaction of DRSP at doses of 3 mg with the cytochrome P450 enzyme mediated metabolism of other drugs is unlikely.

► Pharmacodynamic interaction with antihypertensive medications and non-steroidal anti-inflammatory drugs (NSAID)

Hypertensive women treated with Angeliq and antihypertensive medications, e.g. ACE inhibitors, angiotensin II receptor antagonists, and hydrochlorothiazide may experience an additional decrease in blood pressure.

Combined use of Angeliq and NSAIDs or antihypertensive medications is unlikely to increase serum potassium. Concomitant use of these three types of medications together may cause a small increase in serum potassium, more pronounced in diabetic women.

Other forms of interaction

► Laboratory tests

The use of sex steroids may influence the results of certain laboratory tests, including biochemical parameters of liver, thyroid, adrenal and renal function, plasma levels of (carrier) proteins, e.g. sex hormone binding globulin and lipid/lipoprotein fractions, and parameters of coagulation and fibrinolysis. Changes generally remain within the reference range. Glucose tolerance was not compromised by the use of Angeliq.

4.6 Pregnancy and lactation

Angeliq must not be used during pregnancy and lactation (see section 4.3 Contraindications).

4.6.1 Pregnancy

If pregnancy occurs during medication with Angeliq, treatment must be discontinued immediately.

4.6.2 Lactation

Small amounts of DRSP are excreted with the milk.

4.7 Effects on ability to drive or use machines

No studies on the effects on the ability to drive and use machines have been performed. No effects on ability to drive and use machines have been observed in users of Angeliq.

4.8 Undesirable effects

4.8.1 Summary of the safety profile

Serious undesirable effects associated with the use of hormone replacement therapy are also mentioned in section 4.4 Special warnings and precautions for use.

The most commonly reported adverse drug reactions (ADRs) with Angeliq are breast pain, female genital tract bleeding and gastrointestinal and abdominal pains. They occur in $\geq 6\%$ of users.

Bleeding irregularities usually subside during continued treatment. The frequency of bleeding decreases with the duration of treatment.

Serious adverse reactions are arterial and venous thromboembolic events as well as breast cancer.

4.8.2 Tabulated list of adverse reactions

The frequencies of ADRs reported in clinical trials with Angeliq are summarized in the tables below. The adverse reactions in Table **Angeliq (1 mg E2)** were recorded in 13 phase 2 and phase 3 clinical studies with Angeliq (1 mg E2/ 0.5 mg DRSP; 1 mg E2/ 1 mg DRSP; 1 mg E2/ 2 mg DRSP; 1 mg E2/ 3 mg DRSP; N=2842=100%).

Within each frequency grouping, ADRs are presented in order of decreasing seriousness. Frequencies are defined as very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$) and rare ($\geq 1/10,000$ to $< 1/1000$).

Table: Angeliq (1 mg E2/ 0.5 mg DRSP; 1 mg E2/ 1 mg DRSP; 1 mg E2/ 2 mg DRSP, 1 mg E2/ 3 mg DRSP)

	Angeliq
--	---------

System Organ Class	Very common	Common	Uncommon
Psychiatric disorders		Emotional lability	
Nervous system disorders		Migraine	
Vascular disorders			Venous and arterial thromboembolic events*
Gastrointestinal disorders		Gastrointestinal and abdominal pains	
Reproductive system and breast disorders	Breast pain*** Female genital tract bleeding	Cervical polyp	Breast cancer**

Adverse events in clinical studies were coded using the MedDRA dictionary. Different MedDRA terms representing the same medical phenomenon have been grouped together as single adverse reactions to avoid diluting or obscuring the true effect.

- * Evidence for relatedness and estimated frequency derived from epidemiological studies with Angeliq (EURAS HRT).
'Venous and arterial thromboembolic events' summarizes the following Medical Labeling Groupings:
Peripheral deep venous occlusion, thrombosis and embolism/pulmonary vascular occlusion, thrombosis, embolism and infarction/myocardial infarction/cerebral infarction and stroke not specified as hemorrhagic
- ** Evidence for relatedness derived from post-marketing experience; frequency estimation derived from clinical studies with Angeliq.
- *** Including breast discomfort

For venous and arterial thromboembolic events, breast cancer and migraine see also sections 4.3 Contraindications, 4.4 Special warnings and precautions for use.

4.8.3 Description of selected adverse reactions

Adverse reactions with very low frequency or with delayed onset of symptoms which are considered to be related to the group of continuous combined products for hormone replacement therapy are listed below (see also section 'Special warnings and precautions for use'):

Tumors

- Liver tumors (benign and malignant)
- Sex-steroid influenced malignancies and pre-malignant conditions (if such a condition is known, this constitutes a contraindication for the use of Angeliq)
- Estrogen-only and combined estrogen-progestin HRT has been associated with a slightly increased risk of ovarian cancer in epidemiological studies. The risk may be more relevant with long-term use (several years) (see section 'Special warnings and precautions for use').

Other conditions

- Gallbladder disease (estrogens are known to increase the lithogenicity of the bile)
- Dementia (there is limited evidence from clinical studies with conjugated equine estradiol-containing preparations that hormonal treatment may increase the risk of probable dementia if initiated in women aged 65 years or older. The risk may be decreased if treatment is initiated in the early menopause, as observed in other studies. It is unknown whether these findings also extend to other HRT products.)
- Endometrial cancer (studies have suggested that the appropriate addition of progestogens eliminates the increase in risk resulting from the use of unopposed estrogen)
- Hypertension (Angeliq has the potential to lower blood pressure in women with elevated blood pressure)
- Disturbances of liver function
- Hypertriglyceridemia (increased risk of pancreatitis when using HRT)
- Changes in glucose tolerance or effect on peripheral insulin resistance
- Increase in size of uterine fibroids
- Reactivation of endometriosis
- Prolactinoma (risk of aggravation of hyperprolactinaemia or induction of tumor growth)
- Chloasma
- Jaundice and/or pruritus related to cholestasis

- Occurrence or deterioration of conditions for which association with HRT use is not conclusive: epilepsy; benign breast disease; asthma; porphyria; systemic lupus erythematosus; otosclerosis; chorea minor
- In women with hereditary angioedema exogenous estrogens may induce or exacerbate symptoms of angioedema
- Hypersensitivity (incl. symptoms such as rash and urticaria)

4.9 Overdose

Acute toxicity studies did not indicate a risk of acute adverse effects in case of inadvertent intake of a multiple of the daily therapeutic dose. In clinical studies up to 100 mg of DRSP and estrogen/progestogen preparations containing 4 mg E2 were well tolerated. Overdosage may cause nausea and vomiting and withdrawal bleeding may occur in some women. There is no specific antidote and treatment should be symptomatic.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Angeliq contains 17 β -estradiol, which is chemically and biologically identical to endogenous human E2, and the synthetic progestogen, DRSP. 17 β -estradiol provides hormone replacement during and after the climacteric. The addition of DRSP helps to provide bleeding control and opposes the development of endometrial hyperplasia caused by estrogens.

5.1.1 Effects of estradiol

The loss of the ovarian function, accompanied by a depletion of estrogen and progesterone production, leads to the menopausal syndrome, characterized by vasomotor and organic symptoms. Hormone replacement therapy is indicated to eliminate these complaints.

Of all physiological estrogens, E2 is the most potent one with the highest affinity to the estrogen receptor. Estrogen target organs include, in particular, uterus, hypothalamus, pituitary, vagina, breast, bones (osteoclasts).

Other effects of estrogens include reduction of insulin and blood glucose concentrations local vasoactive effects mediated by receptors, and receptor-independent effects on vascular smooth muscle. Estrogen receptors have been identified in the heart and coronary arteries.

Oral administration of natural estrogens is advantageous in certain cases of hypercholesterolemia in order to maximize beneficial metabolic liver effects on lipids.

After one year of treatment with Angeliq, the mean changes in the HDL-cholesterol levels were small with a slight increase of 1.1% for the combination with 1 mg DRSP and a slight decrease of -1.6% for 2 mg DRSP and -3.4% for the combination with 3 mg DRSP. The serum levels of LDL-cholesterol were on average decreased by -11% (1 mg DRSP), -14% (2 mg DRSP) and -13% (3 mg DRSP) as compared to a -9% decrease after one year of treatment with 1 mg E2 only.

The combinations with DRSP seemed to attenuate the increase in the triglyceride levels caused by treatment with 1 mg E2 alone. After one year of treatment with 1 mg E2, the patients' triglyceride levels were on average approximately 18% above baseline as compared to mean increases of 9% (1 mg DRSP), 5% (2 mg DRSP) and 4% for the combinations of 1 mg E2 with DRSP.

Treatment with Angeliq for two years leads to a mean increase in bone mineral density by about 3 to 5% whereas after placebo treatment a mean decrease of about 0.5% was observed. In both osteopenic and non-osteopenic patients a statistically significant difference in the hipbone mineral density was found between each active treatment and placebo. An increase in bone mineral density in the active treatment groups was also observed in the lumbar spine and the total body.

Long-term treatment with HRT has been shown to reduce the risk of peripheral fractures in non-osteoporotic postmenopausal women.

HRT has also a positive effect on skin collagen and skin thickness and can retard the process of skin wrinkling.

Estrogen monotherapy exerts a dose-dependent stimulating effect on mitosis and proliferation of the endometrium and thus increases the frequency of endometrial hyperplasia and hence the risk of endometrial carcinoma. In order to avoid endometrial hyperplasia, a combination with a progestogen is necessary.

5.1.2 Effects of drospirenone

DRSP exerts pharmacodynamic effects very similar to natural progesterone.

5.1.2.1 Progestogenic activity

DRSP is a strong progestogen with a central inhibitory effect on the hypothalamic-pituitary-gonadal axis. In fertile women, DRSP exerts a contraceptive effect; ovulation is inhibited when DRSP is administered as a monosubstance. The threshold dose of DRSP for ovulation inhibition is 2 mg/d.

Angeliq is a continuous combined HRT given with the intent of avoiding the regular withdrawal bleeding associated with cyclic or sequential HRT. During the first months of treatment bleeding and spotting are quite common but decrease with time. With Angeliq (2 mg DRSP), the amenorrhea rate rapidly increased to 81% already in cycle 6, 86% in cycle 12, and 91% in cycle 24.

DRSP in Angeliq opposes the development of estrogen-induced endometrial hyperplasia effectively. After 12 months of treatment with Angeliq (DRSP doses 0.5, 1, 2, or 3 mg) an atrophic/inactive endometrium was achieved in 71 – 77% of the women.

5.1.2.2 Antimineralocorticoid activity

DRSP has competitive aldosterone antagonistic properties. Blood pressure lowering effects are more pronounced with higher DRSP doses in hypertensive women. Patients with elevated blood pressure treated with Angeliq for 8 weeks experienced a significant decrease in systolic/diastolic blood pressure values (office cuff vs. baseline -12/-9 mm Hg, vs. placebo -3/-4 mm Hg; 24h ambulatory blood pressure measurement vs. baseline -5/-3 mm Hg vs. placebo -3/-2 mm Hg). The effects are apparent within 2 weeks with maximum effects achieved within 6 weeks from the start of the therapy.

Table: Adjusted Mean Change in Systolic/Diastolic Blood Pressure from Baseline Values (mm Hg)						
	Angeliq-1 mg E2/ 2 mg DRSP at Week 8		Angeliq-1 mg E2/ 3 mg DRSP at Week 8		Angeliq-1 mg E2/ 3 mg DRSP at Week 12	
	Office cuff	ABPM	Office cuff	ABPM	Office cuff	ABPM
Change from baseline	-12/-9	-5/-3	-14/-9	-6/-4	-14/-8	-8/-4
Observed reduction corrected for placebo effect	-3/-4	-3/-2	-5/-4	-5/-3	-7/-4	-7/-3

ABPM = 24h Ambulatory Blood Pressure Measurement

Normotensive women did not experience relevant changes in blood pressure.

In clinical trials with Angeliq the mean body weight was either unchanged (1 mg DRSP) or decreased during the treatment period of 12 months by 1.1-1.2 kg (3 or 2 mg DRSP daily) whereas an increase of 0.5 kg was observed in patients treated with E2 alone.

Women who received DRSP in addition to E2 during a clinical trial reported less peripheral edema than those treated with E2 alone.

In women with angina pectoris treatment of 6 weeks with Angeliq (2mg DRSP) improved coronary flow reserve adaptation to stress (relative change + 14% vs. -15% in the placebo group).

5.1.2.3 Antiandrogenic activity

Like natural progesterone, DRSP has antiandrogenic properties.

5.1.2.4 Effects on carbohydrate metabolism

DRSP has no glucocorticoid or antiglucocorticoid activity and has no effect on glucose tolerance and insulin resistance. Glucose tolerance is not compromised by the use of Angeliq.

5.1.2.5 Other properties

Angeliq provides a positive effect on well-being and quality of life. Beneficial effects assessed by the Women's Health Questionnaire were significantly superior to the treatment with E2 alone (total score). This increase was mainly due to improvement in somatic symptoms, anxiety/fears, and cognitive difficulties.

Observational studies and the Women's Health Initiative (WHI) trial on conjugated equine estrogens (CEE) plus medroxyprogesterone acetate (MPA) suggest a reduction of colon cancer morbidity in postmenopausal women taking HRT. In the WHI trial on CEE mono-therapy a risk reduction was not observed. It is unknown whether these findings also extend to other HRT products.

5.2 Pharmacokinetic properties

5.2.1 Drospirenone

5.2.1.1 Absorption

Orally administered DRSP is rapidly and almost completely absorbed. Maximum concentrations of the drug in serum as indicated in the table below are reached about 1 h after single and multiple ingestion of Angeliq. Pharmacokinetics of DRSP are dose-proportional within the dose range of 0.25 to 4 mg. Bioavailability is between 76 and 85%. The intake of food had no influence on the bioavailability of DRSP as compared to drug intake on an empty stomach.

Pharmacokinetic parameter	Angeliq-0.5 mg E2 / 0.25 mg DRSP	Angeliq-0.5 mg E2 / 0.5 mg DRSP	Angeliq-1 mg E2 / 1 mg DRSP	Angeliq-1 mg E2 / 2 mg DRSP*
---------------------------	----------------------------------	---------------------------------	-----------------------------	------------------------------

C _{max} , sd [ng/ml]	3.35	6.05	11.6	21.9
C _{max} , ss [ng/ml]	5.54	10.1	17.6	35.9
AUC(0-24h) sd [ng/ml]	18.5	36.6	82.1	161
AUC(0-24h) ss [ng/ml]	46.6	96.8	194	408

*Data for Angeliq-1 mg E2 + 2mg DRSP were calculated by interpolation between the investigated doses of 1 mg E2 + 1 mg DRSP and 1 mg E2 + 4 mg DRSP .

C_{max}: maximum concentration, sd: single dose, ss: steady-state

5.2.1.2 Distribution

After oral administration, serum DRSP levels decrease in two phases with a mean terminal half-life of about 35 – 39 h. DRSP is bound to serum albumin and does not bind to sex hormone binding globulin (SHBG) or corticoid binding globulin (CBG). Only 3 - 5 % of the total serum drug concentrations are present as free steroid. The mean apparent volume of distribution of DRSP is 3.7 - 4.2 l/kg.

5.2.1.3 Metabolism

DRSP is extensively metabolized after oral administration. The major metabolites in the plasma are the acid form of DRSP, generated by opening of the lactone ring, and the 4,5-dihydro-drospirenone-3-sulfate, formed by reduction and subsequent sulfatation. Drospirenone is also subject to oxidative metabolism catalyzed by CYP 3A4.

5.2.1.4 Elimination

The total clearance of DRSP from serum is 1.2 - 1.5 ml/min/kg. DRSP is excreted only in trace amounts in unchanged form. The metabolites of DRSP are excreted with the feces and urine at an excretion ratio of about 1.2 to 1.4. The half-life of metabolite excretion with the urine and feces is about 40 h.

5.2.1.5 Steady-state conditions

Following repeated daily administration of Angeliq, maximum steady-state concentrations of DRSP in serum are reached as indicated in the table above. Steady-state conditions are reached after about 10 days of daily treatment with Angeliq. Serum DRSP levels accumulated by a factor of about 2 to 3 as a consequence of the ratio of terminal half-life and dosing interval.

5.2.2 Estradiol

5.2.2.1 Absorption

Following oral administration, E2 is rapidly and completely absorbed. During the absorption and the first liver passage, E2 undergoes extensive metabolism, thus reducing the absolute bioavailability of estrogen after oral administration to about 5% of the dose. Maximum concentrations of about 16 or 22 pg/ml were generally reached 2-8 h after single oral administration of Angeliq containing 0.5 or 1mg E2, respectively. The intake of food had no influence on the bioavailability of E2 as compared to drug intake on an empty stomach.

5.2.2.2 Distribution

Following oral administration of Angeliq only gradually changing serum levels of E2 are observed within an administration interval of 24 hours. Because of the large circulating pool of estrogen sulfates and glucuronides on the one hand and the enterohepatic recirculation on the other hand, the terminal half-life of E2 represents a composite parameter that is dependent on all of these processes and is in the range of about 13-20 h after oral administration.

E2 is bound non-specifically to serum albumin and specifically to SHBG. Only about 1-2 % of the circulating E2 is present as free steroid, 40-45 % is bound to SHBG. Orally administered E2 induces the formation of SHBG which influences the distribution with respect to the serum proteins, causing an increase of the SHBG-bound fraction and a decrease in the albumin-bound and unbound fraction indicating non-linearity of the pharmacokinetics of E2 after ingestion of Angeliq. The apparent volume of distribution of E2 after single intravenous administration is about 1 l/kg.

5.2.2.3 Metabolism

E2 is rapidly metabolized, and besides estrone and estrone sulfate, a large number of other metabolites and conjugates are formed. Estrone and estriol are known as pharmacologically active metabolites of E2; only estrone occurs in relevant concentrations in plasma. Estrone reaches about 6-fold higher serum levels than E2. The serum levels of the estrone conjugates are about 26 times higher than the corresponding concentrations of free estrone.

5.2.2.4 Elimination

The metabolic clearance has been found to be about 30 ml/min/kg. The metabolites of E2 are excreted via urine and bile with a half-life of about 1 day.

5.2.2.5 Steady-state conditions

Following daily oral administration of Angeliq, E2 concentrations reached a steady-state after about five days. Serum E2 levels accumulate approx. 2 fold. With a dosing interval of 24 hours, mean steady-state serum levels of E2 fluctuate in the range of 12-29 or 20-43 pg/ml following administration of Angeliq containing 0.5 or 1 mg E2, respectively.

5.2.3 Special Populations

5.2.3.1 Hepatic Impairment

The pharmacokinetics of a single oral dose of 3 mg DRSP in combination with 1 mg E2 was evaluated in 10 female patients with moderate hepatic impairment (Child Pugh B) and 10 healthy female subjects matched for age, weight, and smoking history. Mean serum DRSP concentration-time profiles were comparable in both groups of women during the absorption/distribution phases with similar C_{max} and t_{max} values, suggesting that the rate of absorption was not affected by the hepatic impairment. The mean terminal half-life was prolonged about 1.8 fold and systemic exposure was increased about 2-fold corresponding to an about 50 % decrease in apparent oral clearance (CL/f) in volunteers with moderate hepatic impairment as compared to those with normal liver function. The observed decline in DRSP clearance in volunteers with moderate hepatic impairment compared to normal volunteers did not translate into any apparent difference in terms of serum potassium concentrations between the two groups of volunteers. Even in the presence of diabetes and concomitant treatment with spironolactone (two factors that can predispose a patient to hyperkalemia) an increase in serum potassium concentrations above the upper limit of the normal range was not observed. It can be concluded that DRSP is well tolerated in patients with mild or moderate hepatic impairment (Child Pugh B)

5.2.3.2 Renal Impairment

The effect of renal insufficiency on the pharmacokinetics of DRSP (3 mg daily for 14 days) were investigated in female subjects with normal renal function and mild and moderate renal impairment. At steady-state of DRSP treatment, serum DRSP levels in the group with mild renal impairment (creatinine clearance CL_{cr}, 50-80 ml/min) were comparable to those in the group with normal renal function (CL_{cr}, >80 ml/min). The serum DRSP levels were on average 37% higher in the group with moderate renal impairment (CL_{cr}, 30 - 50 ml/min) compared to those in the group with normal renal function. Linear regression analysis of the DRSP AUC (0-24h) values in relation to the creatinine clearance revealed a 3.5% increase with a 10 ml/min reduction of creatinine clearance. This slight increase is not expected to be of clinical relevance.

5.2.3.3 Ethnic groups

The impact of ethnic factors on the pharmacokinetics of DRSP (1-6 mg) and ethinylestradiol (0.02 mg) was studied after single and repeated daily oral administration to young, healthy Caucasian and Japanese women. The results showed that ethnic differences between Japanese and Caucasian women had no clinically relevant influence on the pharmacokinetics of DRSP and ethinylestradiol.

5.3 Preclinical safety data

Non-clinical data reveal no special risks for humans based on conventional studies of repeated dose toxicity, genotoxicity, carcinogenic potential and toxicity to reproduction. However, it should be born in mind that sex steroids can promote the growth of certain hormone-dependent tissues and tumors.

6. PHARMACEUTICAL PARTICULARS

Nature and contents of container

Calendar-pack containing 28 tablets

Manufacturer

Bayer Weimar GmbH und Co. KG
Dobereinerstrasse 20
99427 Weimar
Germany

Repacker

Bayer AG,
Berlin, Germany

Date of Revision of Text

September 2021